

- iii. **The Indus Delta:** As the Indus River continues its journey towards the Arabian Sea, it becomes very slow and forms its delta near Thatta where it gets divided into many distributaries. The sea tides have changed the coastal areas to 40 kilometers into marshy lands. Therefore, coconut and mangroves are found in this area.

#### 4. The Deserts including Coastal Areas

An extensive area in the south east of Pakistan is covered with rolling sand dunes. The desert areas of Pakistan receive very low rainfall; therefore, these deserts are without natural vegetation. Some plain areas of Pakistan are also called desert or semi desert because their physical conditions are different from those of the plain areas. Some of them are located in the Punjab and some in Sindh Province. These areas are as under:

- i. **Thal Desert:** Thal desert is located in the district of Mianwali, Muzaffargarh and Dera Ghazi Khan. Three fourth of its area is composed of huge sand dunes though most of its areas have been irrigated by Indus canal water.
- ii. **Cholistan Desert:** About sixty percent area of Bahawalpur region and its south eastern area are located in the Cholistan desert. Its main portion is located in India. It is covered with sand dunes and clusters of thorny bushes, acacia etc.
- iii. **Thar and Nara Desert:** The desert area of the Thar and the Nara is located in border areas of Khairpur, south-eastern parts of districts Tharparkar, Umerkot and Sanghar in Sindh. In fact, this area is the extension of Rajasthan desert of India. It is not drained by any perennial stream. Therefore, wind action is dominant in the formation of topography. Vast expanses of sandy plains with innumerable sand dunes dominate the scene. Agriculturally it is a poor area. With the extension of irrigational facilities, some areas have been reclaimed. The natural vegetation of the area is thorny bushes. In the same way, the south western portion of Balochistan is also a desert.



- iv. **The Chaghai and Kharan Desert:** The northwestern districts of Chaghai and Kharan in Balochistan are extremely dry. The average rainfall is not more than 25 mm. These barren areas are most sparsely populated with density of 4 people per kilometer.
- v. **Coastal Area:** Pakistan has about 1050 Kilometers long coastal area. It has two parts. The coast between the border of Iran and Hub River is called Makran coast. Its length is 750 kilometers. The other part is called Sindh coast. Its length is 300 kilometers. It is located between Hub delta and Thatta, Badin district in (Sindh).

All the coastal areas of Pakistan are located on the Arabian Sea. The most important harbour of Pakistan is Karachi. Other ports are Port Qasim, Sonmiani, Ormara, Pasni, Gwadar and Jewani.

### 3. Climate

The long lasting conditions of weather in a particular area such as air pressure, temperature, humidity and rainfall are termed as the climate of an area.

Pakistan is situated in the north of Tropic of Cancer. Pakistan is a sub-tropical country. It is situated in the western part of the monsoon region. Some of the areas in the north of Pakistan are warm and moist. Whereas the mountainous areas have highland type climate.

The average January temperature in the plains of Pakistan is 4°C and maximum temperature of the same month is 24°C, whereas the minimum temperature in the months of June / July is 30°C and maximum temperature in the same months is 48°C. The maximum temperature of Sibi and Jacobabad is about 50°C.

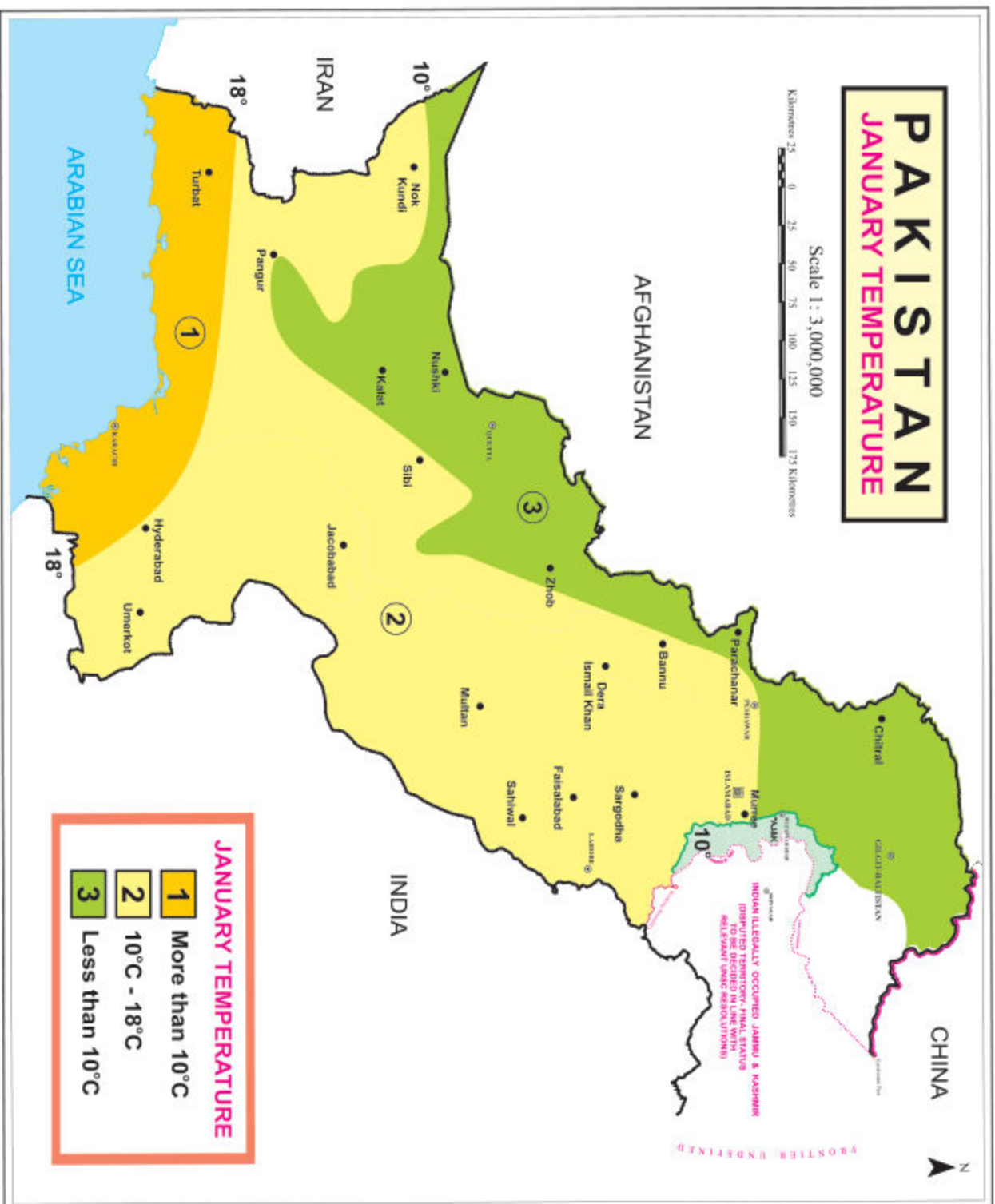
Pakistan is divided into the following regions on the basis of its climate.

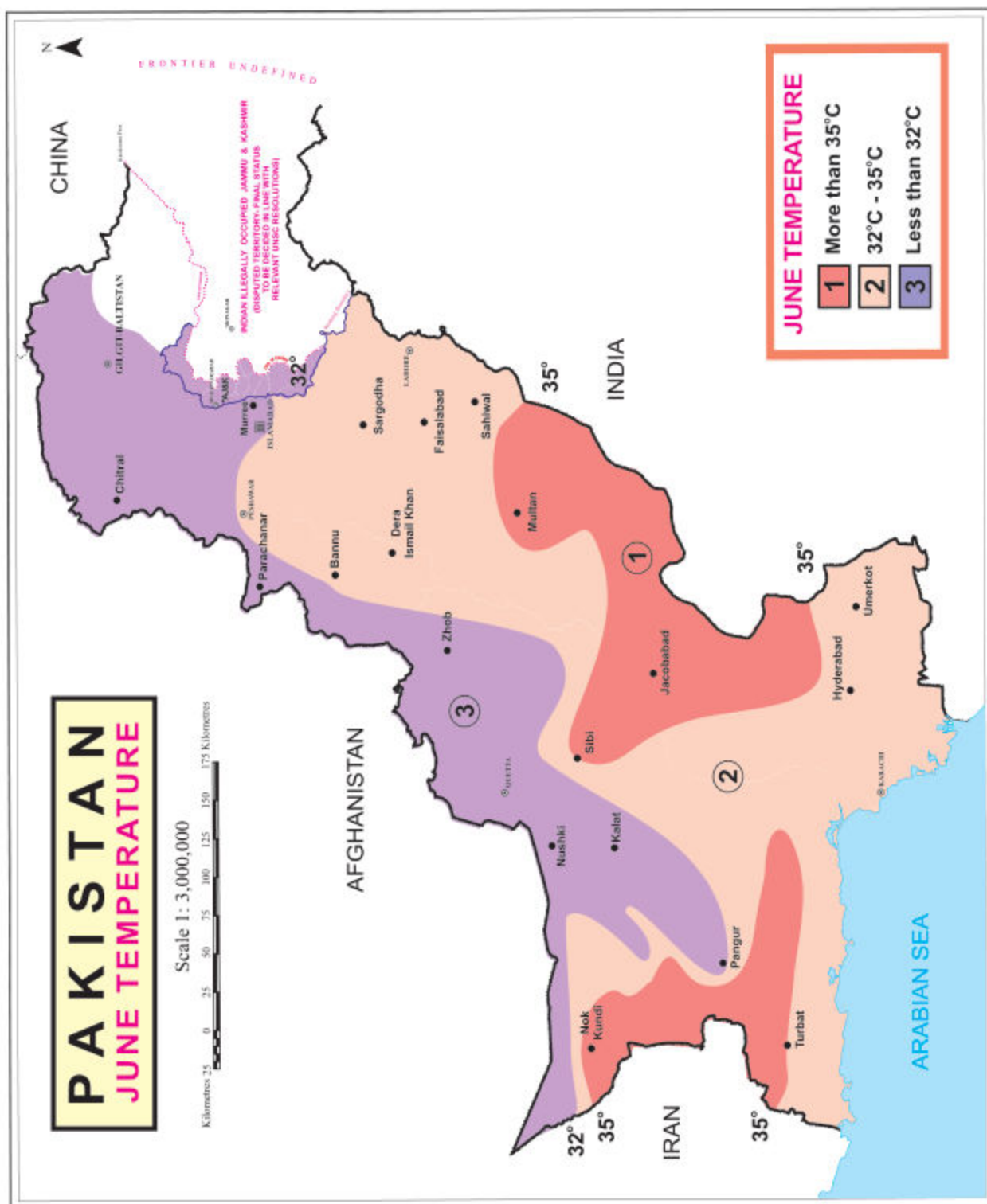
1. Sub-Tropical Continental Highland.
2. Sub-Tropical Continental Plateau.
3. Sub-Tropical Continental Lowland.
4. Sub-Tropical Coastland.

#### **1. Sub-Tropical Continental Highland**

It includes Pakistan's northern mountain ranges (outer and central Himalayas), north western mountain ranges (Waziristan, Zhob and Loralai) and the mountain ranges of Balochistan (Quetta, Sarawan, Central Makran and Jalawan). Here the winter is extremely







cold, normally there is snow fall. Summer season is quite cool and in the spring and winter seasons, there is mostly fog and rainfall.

In some areas of this region like outer Himalayas, Muree and Hazara districts, rainfall continues throughout the year. Mostly it rains at the end of the summer season.

## **2. Sub-Tropical Continental Plateau**

In this climatic region most of the parts of Balochistan are included. From May to the mid of September hot and dusty winds continuously blow. Sibi and Jacobabad are located in this region. There are a few mm rainfalls during the months of January and February (normally 50 mm). Extreme hot, dry and dusty winds are important characteristics of this region.

## **3. Sub-Tropical Continental Lowland**

This climate region includes the upper Indus plain (Punjab province) and lower Indus plain (Sindh province). Summer is very hot. The north of Punjab receives late summer monsoon rains. While the rest of Punjab receives less rainfall. Winter rainfall situation is same. In the Thal, Kachhi, Sibi and South Eastern plains are dry and a large area is a desert. The Peshawar plain experiences thunder storms and dust storms during summer.

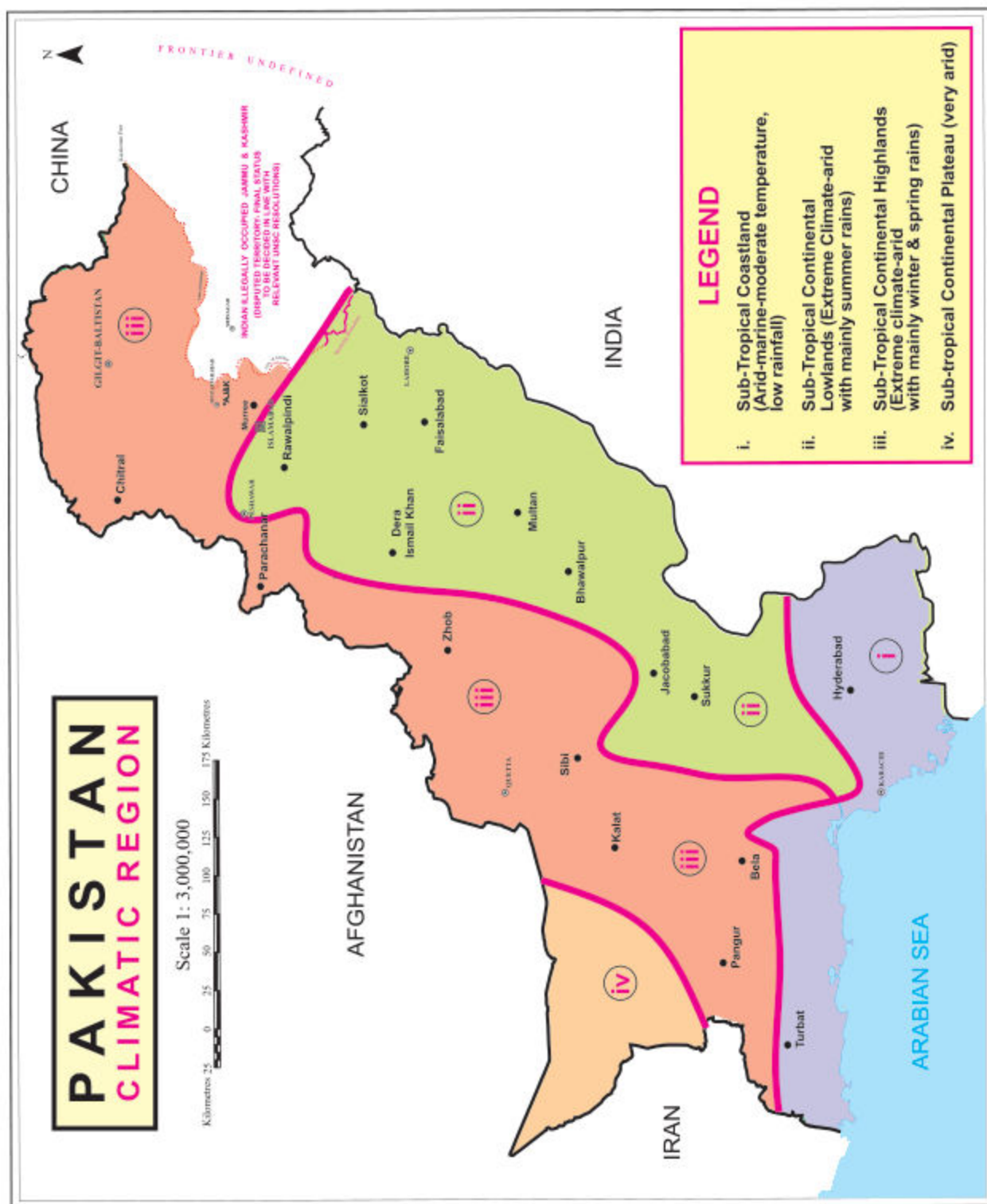
## **4. Sub-Tropical Coastland**

This climatic region includes the coastal areas of Sindh and Balochistan provinces, or the southern part of Pakistan. The temperature is moderate. The difference between maximum and minimum temperature or range of temperature is less. There is inflow of sea breeze throughout the summer and the humidity is high.

Annual average temperature is 32°C and rainfall is 180 mm. May and June are the hottest months. The plain of Lasbela receives rainfall both in summer and winter seasons.

Although Pakistan is situated in the monsoon zone of climate, but being in the extreme western part of this region, does not possess its characteristics. Therefore, the climate of Pakistan is arid, hot and continental type. There are extreme variations in the temperature. A major part of Pakistan lies at a great distance from the sea.





## **Influence of Climate on life**

Human life is greatly affected by the influence of climate. The climate and weather conditions of a particular area influence the way of living, dress, food, occupation, sports, customs and the economic set up of the people of that area. Pakistan is a vast country and there is much difference of climate in its various regions. There is a marked difference in the customs and living patterns of the people.

In northern mountainous areas of Pakistan, the winter is extremely cold. The temperature falls below freezing point and snow spreads all around. Severe cold, influences human, animal and the plant life of the area. The people of this area start storing necessary food items and the animals before the start of winter. During winter people are confined to their houses. The people pay attention to household cottage industry. Some people of this area temporarily migrate to plain areas during winter in search of jobs and food and return back to their homes when summer season starts. At the start of the summer season, when snow melts, life generates through hustle and bustle. The trees, plants and grass grow rapidly. Springs and streams, which had become frozen during winter, start flowing with clean water and become a source of adding beauty to the areas. People start outdoor activities like cultivation, trade, labour etc. The cold climate has an effect on the health of the people of Northern areas. They possess sound physique. Their colour is fair. They are hardworking and brave. The tough life patterns have made them strong and courageous.

The plains of Pakistan have severe hot season during summer and fairly cold during winter. During winter it is possible to work diligently but during summer the working efficiency decreases. Light clothes are used during summer and heavy woolen clothes are used during winter. The land and climate of these areas is suitable for agriculture. Variety of crops is grown during the winter and the summer. The area becomes prosperous as a lot of food grain, vegetables and fruit grow in these areas. The plain areas are thickly populated as compared to the northern areas. Means of transportation are convenient. People are directly or indirectly associated with agriculture. Educational and other amenities of life are available. People have many jobs to do.

Most of the southern areas of Pakistan are desert and hot. Windstorms and sandy cyclones blow frequently. People use coarse clothes to protect them from heat and hot wind. They put on big turban and cover their bodies with clothes. They travel in the night because deserts in the night are cool. The people breed sheep, goats and cattle. The areas irrigated by canals are under cultivation.



The Balochistan Plateau have extreme climate. Winter is cold and in some areas there is snowfall. During winter people are engaged in indoor activities and most of the time people prepare gifts for sale. Many people of cold areas migrate to warm areas and return back in summer. In summer the plains of Balochistan are very hot. People use loose clothes. Water is collected through underground artificial streams called *KAREZ*, now mostly dried up. Now a few dams are being constructed to provide water for irrigation and drinking.

In short, people living in severe cold region wear woolen and thick clothes. The people of cold region build houses comprising of small rooms which can be heated easily. The people living in cold regions consume food with high protein and fat to maintain the temperature. They use fatty meat, eat wheat and corn bread. They drink tea and coffee. Movement in the cold regions is not frequent and easy. During snowfall roads are closed. People are confined to their homes. Population in cold region is thin. Winter is dull but life is pleasant during short summer. Job opportunities are limited. The people of these regions are not much prosperous.

As compared to cold region, the people living in low lands and desert region put on loose dress due to hot weather in summer. Their houses are open and airy. People of hot regions eat wheat bread, rice and fish. They drink syrups of different kinds. People are involved in farming and cultivation throughout the year. People perform different types of jobs which include trade, business and employment in government offices and in private organizations as well. Low land regions are thickly populated due to job opportunities and availability of amenities of life. Life is full of activities. People are prosperous as compared to the people of cold regions.

## 4. Environmental Problems

Environmental pollution has emerged as a big problem throughout the world. This is due to increasing population, rapid industrialization and due to the smoke created by heavy and light vehicles. Ecology has emerged as a new science to make people aware about the pollution of water, air and land. Pakistan is among those countries of the world where environmental pollution is increasing day by day.

### **Environmental Pollution**

Pollution means such changes in the environment of earth, water and air, which put injurious influence on human life as well as on the animals. When different elements bring unhealthy changes in our environments, it is called environmental pollution. This environmental pollution is of three types. They are:



- i. Air Pollution (Atmospheric)
- ii. Water Pollution
- iii. Land Pollution

**i. Air Pollution (Atmospheric)**

Neat and clean atmosphere is essential for human health. Air is the most essential element for human survival. There are a number of gases in the air. But air is basically composed of Oxygen, Nitrogen and Carbon dioxide. The water vapour is about 4% in the air atmosphere like soot, smoke, Carbon dioxide and Carbon mono oxide making the environment polluted. Gases with injurious effects are also present in the atmosphere. An adult person needs about 15 kilograms of air every day. The amounts of gases, which are consumed, become part of atmosphere again. This maintains the ratio of gases in the atmosphere. But natural balance of these gases is rapidly changing due to a number of human activities. The smoke, poisonous moisture, atomic wastes etc. are damaging the atmosphere. This is called "Atmospheric pollution" This pollution brings unhealthy impact on human beings, vegetation and the buildings. Man has invented machines for his convenience and for living a comfortable life. The oil, gas and coal are used to run motor cars, aeroplanes and ships. They produce dark smoke, which is injurious to human health. Industrial development has made human life comfortable but on the other hand this material development has increased air pollution. With the growth of industrial development, air pollution has also increased. It is, therefore, necessary to counter the root causes of this pollution.



## ii. Water Pollution

Water is a great blessing of Almighty Allah and is an important need of human beings. Water maintains 70% of the total weight of human body. The total area of the planet Earth is 71% water and 29% land. The potable water of the globe is 2.8% of the total quantity of water on the earth. This fresh water is used in the homes, in agriculture and factories. The 0.65% of this water is in liquid form and the rest is in the form of snow and glaciers. The remaining 97.2% water on the earth is salty and not suitable for drinking and for other use. Water is a good solvent; therefore, it dissolves solids, gases and liquids. When unclean bodies are mixed in water, it becomes dense and loses its natural form. This state of water is called "Water pollution". The polluted water is not only harmful for human health but also for animals, agriculture and vegetation. Polluted water becomes source of dangerous diseases and sometimes causes death.

## iii. Land Pollution

The earth is polluted biologically, chemically and physiologically, when a number of harmful things are inducted in the earth. Air and water pollution are also the source of land pollution. Land is polluted through floods, earthquakes, volcanoes and fire. When human beings bury harmful material into the earth it damages its composition. Elements which make the land barren are as under:

1. Decrease in the forest areas and stock rearing of bushes and low height trees.
2. Salinity and water logging.
3. Depositing waste of pharmaceuticals and chemicals into the earth.
4. Excessive use of fertilizers.
5. Water and air pollution.





## 5. What should we do now?

In order to prevent and control pollution, a number of Conferences have been held in the World. Atmospheric temperature and air pollution is increasing day by day but little attention is being shown to this problem. The reason is that the big industrial countries are not inclined to gradually decrease the release of injurious gasses of their industries. Because their industrial output will be affected.

As far as Pakistan is concerned the following arrangements will help in reducing environmental pollution:

- i. The poisonous chemical material should be separated and cleaned before it is drained into the rivers or their deposition into the ditches.
- ii. The dirty water (sewerage) of the cities should be cleaned through machines in big ponds as is done in the developed countries before it is drained into the rivers.
- iii. Vehicles which emit frequent smoke should be banned. Those who violate it should be fined. The garbage should be burnt in an appropriate manner.
- iv. The role of the institutions involved in the environmental affairs should be strengthened including provision of adequate funds for better environment.
- v. The human and animal waste should not be deposited in the fields nor should it be used as fertilizer.
- vi. Arrangements for more plantations should be made. The cutting of forests should be stopped because these forests are the best source of purifying the atmosphere.
- vii. The use of polythene bags should be stopped. If it is not possible, awareness campaign should be started advising the people not to throw these bags in open places.
- viii. Arrangements for general cleanliness should be made around the public places, parks and streets.
- ix. Radio and television programmes on environmental pollution should create awareness among the people about the dangers of environmental pollution.

## EXERCISE

### A. Answer the following questions:

1. What do you understand by Climate?
2. What are different climatic regions of Pakistan?
3. How does climate affect human life?
4. What is Environmental Pollution?
5. What is the relationship between the population growth and the environmental pollution?
6. Describe the importance of the location of Pakistan in the South Asian Region.
7. Give a brief account of the physical features of Pakistan.
8. Mention the benefits of the Northern-eastern mountainous region.
9. Name the neighbouring countries of Pakistan.

### B. Fill in the blanks:

- i. The long border between Pakistan and Afghanistan is called \_\_\_\_\_.
- ii. Pakistan is divided into \_\_\_\_\_ climatic Regions.
- iii. The coastal areas of Pakistan are \_\_\_\_\_ and \_\_\_\_\_.
- iv. The climate of the upper Indus plains of Pakistan is \_\_\_\_\_.
- v. The water evaporation is about \_\_\_\_\_ in the atmosphere.
- vi. Water maintains \_\_\_\_\_ of the total weight of human body.
- vii. People of the Northern region in Pakistan perform \_\_\_\_\_ activities during winter.
- viii. In the South-West of Pakistan, \_\_\_\_\_ neighbouring country is our neighbouring country.
- ix. Balochistan has an area of \_\_\_\_\_ sq. kilometers.
- x. The word FATA means \_\_\_\_\_.
- xi. Hub and Lyari Rivers flow in \_\_\_\_\_ range of mountains.
- xii. The biggest salt mine in the world is in Pakistan's \_\_\_\_\_ Province.



# Chapter 05

## RESOURCES OF PAKISTAN

### 1. Resources

There are two types of resources. One is the human resource, which means capacity and capability of human beings for doing certain works. The types of professions differentiate the human beings from each other. All these professions combined together are known as human resources. The other type of resource is the natural resource provided by the Nature. The natural resources are the source of production. The two categories of human and natural resources are mentioned as under:-

#### i. Human Resources

People engaged in so many occupations and tasks constitute human resources. These resources combine the capacity, capability, and expertise of the human beings working in different professions. These human resources are also called the manpower of a country. This manpower has been categorized in different employments. It may be noted that a person who is above 10 years of age and works for himself or performs duty for another for at least one hour a day is called an employed person. Pakistan's manpower is associated with different professions and occupations, like agriculture, mining, architecture, business, communication, government service and other types of paid work.

#### ii. Natural Resources

These are the resources which the Almighty Allah has given to different countries of the world in the form of soil, forest, mineral, water, etc. These resources are the free gift from Allah. Man can explore them and get benefit out of these natural resources."



## Importance of Resources in National Development

Resources are important because:

- i) They are the real wealth of a nation. Those countries where human or natural resources are in plenty have shown tremendous progress and prosperity. However, progress and prosperity depends on the wise and appropriate utilization of these resources. There are a number of countries in the world where both human and natural resources are available but due to the lack of planning and hard work the benefits from these resources are quite meager.
- ii) Human and material resources help in building strong defence for the country. All material requirements for strong military build-up are provided through natural resources. Human resources help utilize these resources.
- iii) They are a source of high prestige for a country. For example, all western countries attract the people of developing countries to take advantage from the rich resources of the developed countries.
- iv) They help in meeting the material requirements of the country which include necessities, comforts and luxuries of life.
- v) They help in flourishing trade and business. Those countries which are rich in resources have captured the world trade. Their economy is strong.
- vi) They provide employment opportunities to the people. It is because of these resources that millions of people go to Gulf countries and Middle East countries for employment. Similarly, European countries, USA, Canada and Australia have the attraction for better employment opportunities.
- vii) They help in rapid progress and development of a country.
- viii) They provide people with ample food for living and other amenities of life. They help in building national institutions. They strengthen national unity and individual character. They help in developing virtues like honesty, truthfulness and generosity. The obvious reason is that people of developed countries have no fear of losing livelihood or facing financial set-back.



## 2. Natural Resources

Pakistan is a gift of Allah with plenty of natural resources. These natural resources are as under:

### A. Soil

Soil has been named as part of unconsolidated material which covers the surface of earth and becomes a source for growth of plants. Generally soil is a compound of salt, silt and clay. The soil of Pakistan differs from area to area. Generally soil in Pakistan is of red, white and black colours. The particles present in the soil determine its colour. For example, presence of iron particles in the soil makes its colour red. There are three layers of soil. The upper layer helps growing vegetation and plants. The soil created as a result of erosion of rocks and climatic changes is called local soil. The third soil is called the transferred soil because it changes the place due to natural consequences. There is a variety of soil in different parts of Pakistan. The following kinds of soil are found indifferent parts of Pakistan:-

#### i. Indus Basin Soils

The Indus Basin soils are created by depositing the alluvium by the river Indus and are generally called Indus Basin Soils. These can further be divided into three major groups. First group is known as Bangar Soil of alluvium which is very productive when irrigated and fertilized. This soil covers a vast area in the Indus plain, including most of the area of the Punjab, Peshawar, Mardan, Bannu and the greater part of the Indus Plain. The second group is called Khaddar Soil and is formed from the flooded layers of silt, loam and silty clay loam. Such type of soil is commonly found in Mardan and Bahawalpur. This soil is very fruitful for agriculture products when plenty of water is added into it. The third group of soil is Indus Delta Soil covering the whole of Indus Delta from south of Hyderabad to the Arabian Sea coast. Before becoming part of the sea, Indus River is distributed here into a number of branches. About one third of the area is covered by clay soil which is developed under flooded water conditions. This soil is used for the cultivation of rice.



## **ii. Mountain Soils**

North and North Western Mountain's Soil are brown. These soils are residual as well as transported under arid and semi-arid conditions. In the mountain valleys, soil is formed from the alluvial deposits of the streams. These soils are calcareous silt loam and sandy loams of organic content. In the sub-mountain area of Potwar. Plateau shallow residual soils are found. These soils become productive with plenty of water. The lowest part of the inter-mountain valleys and interior basins are arid or semi-arid regions. The soils of these areas are saline. The colour of soil is reddish. Most part of this soil is barren.

## **iii. Sandy Desert Soils**

Sandy Desert Soils extend over some areas of Western Balochistan, Cholistan and Thar Desert. These soils are of yellow colour, which contain phosphate, iron, lime and potash. These are essential for the fertility of land.

## **B. Forests**

Forests are an integral part of a country's economy. For balanced economy of a country about 25% of its land should be under forests. Forests are great source of natural resources. In Pakistan only 4.8% land is under forests. Provincial distribution in Pakistan indicates 2.7% land under forests in Punjab, 4.2% in Sindh, 15.6% in Khyber Pakhtunkhwa, and 2.1% in Balochistan.

The climate of Pakistan is too dry for growing forests except in the northern hilly area and sub-mountain hills, where forests are due to sufficient rainfall and hill slopes. In Pakistan the forest area is being reduced due to ruthless cutting of forests, use of land for construction of houses and due to rivers erosion each year. It is necessary that more land is put under forests and un-necessary wood cutting is stopped.

Area wise and on the basis of climate, forests in Pakistan are divided into six kinds. They are:

- i. Mountainous Forests:** These forests are found in North and North Western Mountain area. These areas include Swat, Dir, Chitral, Abbottabad, Murree and Mansehra. These are evergreen coniferous soft wood forests because annual rainfall in these areas is more than 100 centimeters. The principal coniferous trees include fir, deodar, blue pine and spruce. They grow generally on a height 1000 to 4000 meters above the sea level. Below 1000 meters height grow trees like oak, maple, birch, walnut, chestnut, mulberry, apple and other fruit trees. These trees are a good source of timber and fruits.



- ii. **Sub-mountainous Forests:** These forests grow at a height of 1000 meters above the sea level. These are found in Kohat, Mardan, Rawalpindi, Attock, Gujrat and Jhelum districts. The most popular trees include phulahi, kahu, jand, shisham, poplar and blackberry. These are hard wood trees and used for fire and for construction work.
- iii. **Dry Western Mountainous Forests:** The trees like chilghoza, pine and juniper grow at high altitude. In other areas, only trees and small thorny bushes grow in Quetta, Kalat, Zhob and Ziarat.
- iv. **Riverine Forests:** These forests grow along the rivers. Trees like shisham, acacia and mulberry are found in these forests. These forests are also found in the canal areas of Punjab and Sindh.
- v. **Canal Forests:** These forests have been planted where canal water is available in abundance. These areas are Chhanga Manga, Chichawatni, Khanewal, Thal, Shorekot, Bahawalpur, Sukkur, Taunsa, Guddu and Kotri. The most popular trees in these forests are shisham, mulberry and eucalyptus.
- vi. **Coastal Forests:** In the South of Sindh near the coast of Arabian Sea, there are some Mangrove forests. The Mangrove forests are breeding grounds for fish and shrimps.



## Advantages of Forests

Following are the advantages of forests:

- i. The forests are one of the important resources for the country and meet a number of requirements like wood, timber, herbs etc.
- ii. The forests help in reducing water logging and salinity and help in restoring fertility of the land.
- iii. The forests moderate the temperature and make the weather pleasant.
- iv. The forests provide herbs which are used in the preparation of medicines.
- v. The forests are the source of wild life. A number of wild animals like lion, leopard, bear, etc., are found in the forests.
- vi. The forests are the main source of firewood.
- vii. The forests add to the beauty of landscape.
- viii. The forests provide a number of resources. For example, the wood from the forest is used for making furniture, paper, matchsticks, sports goods etc.
- ix. The forests protect both human beings and natural vegetation from the onslaught of violent windstorms.
- x. The forests stop quick melting of snow on the mountains. They check soil erosion.
- xi. The forests do not allow the excess of carbon dioxide gas in the atmosphere because they need this gas to take it and release oxygen, necessary for human life.
- xii. The forests are the natural pastures. Animals like goats, sheep and camel get their food from forests.
- xiii. The forests serve as excursion resorts and provide recreation for the people.
- xiv. The forests are the source of growth and breeding of different kinds of animals and birds.



## C. Minerals

Minerals are the natural wealth buried underneath the earth. Almighty Allah has given lot of mineral wealth to Pakistan. These mineral resources are playing a vital role in the rapid economic and industrial progress of the country. Following are the important mineral resources of Pakistan:



### **i. Mineral Oil**

Mineral oil is an important wealth of the modern age. It is a source of generating energy. Mineral oil is found in raw condition which is refined in the oil refineries to get petrol and other by-products like kerosene oil, diesel, plastic, candle, baseline etc.

Pakistan produces about 20% oil of the total requirements of the country. The remaining 85% requirement is met from the import of oil from other countries.

The oil resources of Pakistan are located in Potwar Plateau, Khor, Dhullian, Kot Mayyal, Sarang in Attock district, Balkasar in Chakwal district, and Joyamir in district Jhelum, Dhodhak in Dera Ghazi Khan. In Sindh, oil has been discovered at Badin, Hyderabad, Sanghar, Dadu, Jamshoro, Tande Mohd Khan, Tando Allayar, Matiari, Khairpur, Kashmore, and Ghotki districts.

Oil and Gas Development Corporation (OGDC) is responsible for the exploration of oil in the country. It is continuing its efforts to find more oil resources.



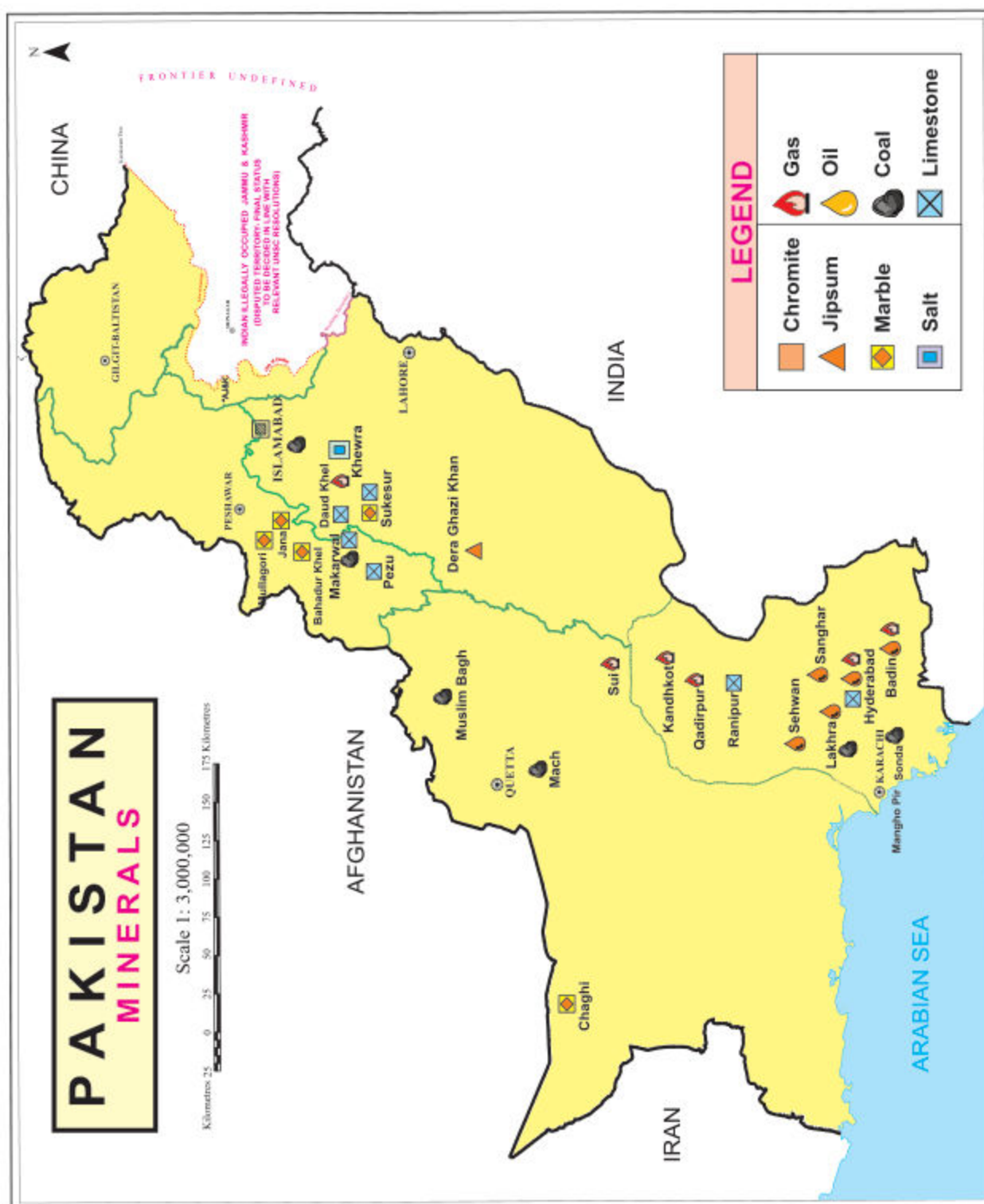
### **ii. Natural Gas**

Natural gas is required for running industry and is used as fuel for the vehicles and household purposes. Its use has become common because it is cheaper than the petrol. About 35% of our energy needs are met out of natural gas. Pakistan has vast reserves of gas. Natural gas was first discovered at a place called Sui near Dera Bugti in Balochistan in 1952. Afterwards, this gas was discovered at thirteen different places in the Lower Indus Valley and Potwar. The most important places for gas reserves are Sui, Uch, Zin (Balochistan), Haripur, Mazrani, Seri, Hundi, Kandhkot, (Sindh), Dhodhak, Pir Koh, Dhullian and Mayyal (Punjab). Natural gas has been brought to different big cities through pipelines. The gas is used in cement, fertilizer and general industries. It is also used for producing thermal electricity.

### **iii. Coal**

Coal is found at a number of places in Pakistan but neither it is of good quality nor it meets the total requirements of the country. Pakistan produces about 65% of coal requirement of the country. In Punjab, the coal mines are at Dandot, Makkarwal and Pidh. The coal mines of Makkarwal in Mianwali district are the biggest coal mines in the country. In Balochistan, coal has been discovered at Sharag, Khost, Hamai, Sar,







Degari, Shirin and Mach. Coal mines In Sindh, coal mines are at Jhimpir in Thatta and Lakhra in Jamshoro district. Coal has also been discovered in Tharparkar district (Sindh) in a vast quantity.

**iv. Iron ore**

Iron ore is very important mineral. It is used for making iron, steel, machines and tools of different kinds. The largest deposits of iron ore have been discovered in the Kala Bagh area. Other deposits are found in Langrial, 32 km south of Abbottabad in Hazara district and Chitral. In Balochistan Iron Ore is found at Khuzdar, Chilghazi and Muslim Bagh. The iron found in Pakistan is not of good quality and it meets only 16% of the total requirement of the country. Pakistan Steel Mill is using imported iron ore.

**v. Chromite**

It is a white colour metal used for making steel, aeroplanes dyes and photography items. Pakistan has the largest reserves of chromite in the world. Most of which is exported to earn foreign exchange. Its deposits are found in Muslim Bagh, Chaghai, Kharan (Balochistan), Malakand, Mohmand Agency and North Waziristan (Khyber Pakhtunkhwa and FATA).

**vi. Copper**

Copper is used for making electrical goods. Electric wire is made out of copper. Balochistan has rich reserves of copper. The copper deposits are at Saindak in Chaghai district of Balochistan.

**vii. Gypsum**

It is a white colour shining stone. It is used in the production of cement, chemical fertilizers, plaster of Paris and bleaching powder. The gypsum is obtained from the districts of Jhelum, Mianwali, Dera Ghazi Khan (Punjab), Kohat Khyber Pakhtunkhwa, Rohri (Sindh) and Quetta, Sibi, Loralai (Balochistan).

**viii. Salt**

Pakistan has the largest reserves of mineral salt in the world. The salt range is located in the south of Potwar Plateau. This salt is of fine quality. The biggest salt mine is in Khewra (Jhelum district). Salt is also obtained from Warchha (Khushab district), Kala Bagh (Mianwali district) and Bahadur Khel (Karak district). In the Mauripur area near Karachi and the Makran coast salt is also obtained from sea water.



#### ix. Limestone

It is mostly used for making cement. When it is burnt, lime is obtained which is used for white washing the houses. It is also used in glass, soap, paper and dye industries. Vast limestone deposits are found in Dandot (Jhelum district), Zinda Pir (Dera Ghazi Khan), Moghul Kot and Ganja Takkar near Hyderabad, Manghopir, Kot Diji and Ranipur (Sindh).

#### x. Marble

Marble of different kinds and of different colours is found in abundance in Pakistan. It is found in the districts of Chaghai, Mardan, Swat and Khyber Agency. It is considered to be of the best quality in the world due to its softness, beauty and colour. Black and white marble is found in the Kala Chitta Hills in Attock district. Pakistan's marble industry is earning huge foreign exchange for the country through export of things made of marble.



### 3. Agriculture

The upper and lower Indus plains have been the centers of human civilization and culture since olden times due to their fertility. Many kinds of crops, fruits and vegetables are grown in these areas. Agriculture sector produces about 30% of the total raw materials and is a source of providing employment to about 55% of population. Earning from agriculture export is 70%. Pakistan is self-sufficient in the production of wheat, rice, cotton and sugar.

#### A) System of Agriculture

Pakistan grows two major crops during the year. The crop grown in October and November is called Rabi crop. It is harvested in the month of





April and May. Wheat, barley, gram and oilseeds are Rabi crops. The second crop is known as Kharif crop. It is grown in the month of May/June and harvested in September and October. Kharif crops are rice, maize, cotton, sugarcane, sorghum jowar) and millet (bajra). These two types of crops are further divided into two kinds. One kind is called food crops, which are wheat, rice, pulses, maize etc. The other kind is called cash crops, which are a big source of earning foreign exchange by selling them into international markets. These crops are cotton, sugar, tobacco etc.

Agriculture occupies an important place in our economic system. Pakistan's economic system is very much linked with agriculture. The main purpose of our economic policy is to make the country self-sufficient in food and other agricultural products. Therefore, all efforts are directed towards attaining self-sufficiency in food. This will help in saving foreign exchange spent on the import of food grains and oilseeds.

Pakistan is following feudal system. The agriculture sector is dominated by the feudal lords who have vast areas of land. The land is thus cultivated through the peasant system. In order to remove the defects of agriculture, several agricultural reforms have been made. Through these reforms millions of acres of land were acquired from the landlords and sole to tenants to intensify the use of this land. Efforts were made to strengthen better relations between the landlords and tenants and protect the rights of both the parties. To promote mechanized farming small scattered pieces of land were joined together. Loans are granted to cultivators for purchasing modern agricultural equipment's, fertilizers, seeds and pesticides.

## Agriculture Products

The crops of Rabi and Kharif are divided into two parts, namely, cash crops and food crops.

### Cash Crops

These crops are the main source of earning foreign exchange. These are:

#### a. Cotton

It is the most important cash crop of Pakistan and a source of uplifting economy of the country. Cotton is called the silver fiber of







Pakistan. Cotton is mainly sown in Punjab and Sindh Province. In Balochistan and Khyber Pakhtunkhwa cotton is sown on a limited scale in a few areas. Two types of cotton are sown in Pakistan one is indigenous and the other is American Cotton. The American cotton has long staple and more attention is given to its cultivation. Due to plenty of cotton produce a number of textile mills have been established in the country. These textile mills produce a variety of fine cotton cloth, cotton yarn and other cotton-based products.

**b. Sugarcane**

It is also an important cash crop, which is sown in all the four provinces of Pakistan but the main areas of its production are the province of Punjab, Sindh and Khyber Pakhtunkhwa. Sugarcane is the source of producing sugar. Its residue is used for making paper. Pakistan produces surplus sugar, which is exported to earn valuable foreign exchange.

**c. Tobacco**

It is another cash crop of Pakistan. Tobacco is mainly produced in the Khyber Pakhtunkhwa, especially in the Peshawar and Mardan districts. It is used for preparing cigarettes. It is also used in cigars. A few big cigarette manufacturing factories are working in the country. Tobacco and its products are also exported to other countries.

**d. Oil seeds**

A variety of oilseeds are produced in Pakistan. The cotton seed, which is the by-product of cotton, is the most important seed. Other oilseeds are rapeseeds, mustard, groundnuts, sesame (sesamum indicum), Lin seed and sunflower. Most of these seeds are produced in the irrigated lands. Our vegetable oil industry is the user of these seeds. But production of oil seeds has not matched with the needs of the country which are imported from other countries.





## Food Crops

These are the crops which are a source of providing food for the people. These food crops are as under:

### a. **Wheat**

It is the staple food grain of Pakistan. Flour is the main by-product of wheat. Bread and other food items are prepared from flour. Three-fourth (3/4) of the total production of wheat is received from the Province of Punjab. After Punjab the province of Sindh produces plenty of wheat. Balochistan and Khyber Pakhtunkhwa are also producer of wheat but their yield is not as substantial as that of the Punjab and Sindh. Pakistan's self-sufficiency in wheat depends on the quantity of water. When nature is benign we produce surplus wheat. Sometimes wheat is imported from other countries. In the year (2004) a large quantity of wheat was imported from other countries. Wheat is an important element of our daily food.

### b. **Rice**

It is the second most important cereal of Pakistan. Pakistan is self-sufficient in the production of fine quality rice (Basmati), which is also exported to other countries. Rice is grown in the canal areas of Punjab and Sindh because lot of water is required for its growth. Gujranwala, Sialkot, Sheikhupura, Sargodha and Sahiwal are very important for rice production. In Sindh, Sukkur, Shikarpur, Larkana and Dadu districts are famous for rice production. Rice is an important food item of the people of Punjab and Sindh. Rice is also grown in a few areas of Khyber Pakhtunkhwa.

### c. **Maize (Corn)**

It is a food crop but is also used as animal fodder. It is mostly grown in the Khyber Pakhtunkhwa, where Mardan, Abbottabad, Mansehra, Swat and Peshawar are noteworthy. In the Punjab, Faisalabad and Sahiwal districts are known for maize cultivation.



**d. Sorghum and Millets (Jowar and Bajra)**

Sorghum and Millets are cultivated to yield food grains and also for green or dried grass to feed mainly milk cattle. These are winter (Kharif) crops, which can be cultivated, even in the areas where soil is poor and prone to drought. Its cultivation is confined to Punjab and Sindh provinces. In the Punjab; Attock, Gujrat, Sialkot and Sargodha districts grow millet. Tharparkar, Umerkot and Mirpurkhas districts in Sindh lead Pakistan in the production of millet. Sorghum is also cultivated in the Northern districts of Punjab i.e. Attock, Rawalpindi, Jhelum and Sargodha. In Sindh, Sukkur, Khairpur, Nawabshah, Naushahro Feroze, Sanghar and Dadu districts are the main areas of its cultivation.

**e. Pulses**

Different varieties of pulses are also cultivated in the country. The gram leads the list of pulses. The rain-fed (Barani) areas of Mianwali and Sargodha are its important centres. In the Khyber Pakhtunkhwa gram is grown in good quantity in Bannu and Dera Ismail Khan Districts. Other varieties of pulses such as Lathyrus (Moong), Lentils (Masoor) and Vetchling (Mash or Urad) are grown more in Punjab than in other parts of the country.

**f. Barley**

It is not grown in large areas. It is grown on a limited scale in less fertile and dry areas of the country. It is normally used by poor families. It is also used as fodder for animals. It is not popular in the country.

**g. Vegetable and Fruits**

Vegetables are grown locally according to the needs of the area all over the country. The important vegetables are potato, pumpkin, tomato, ladyfinger, brinjal, spinach, onion, radish, pea, sugar beer, cabbage, carrot etc. Pakistan is self-sufficient in the production of vegetables. Pakistan exports onion and potatoes to other countries.

